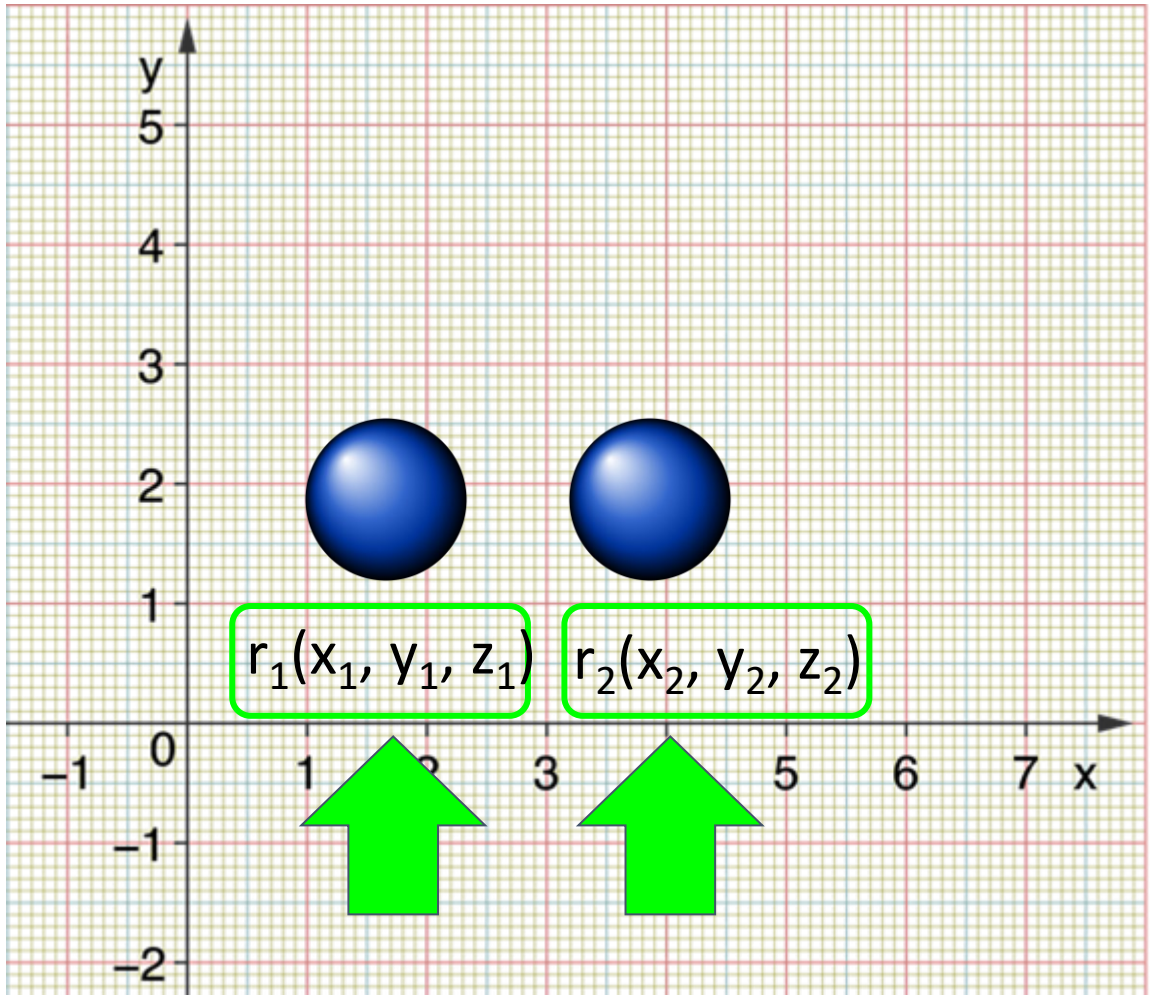
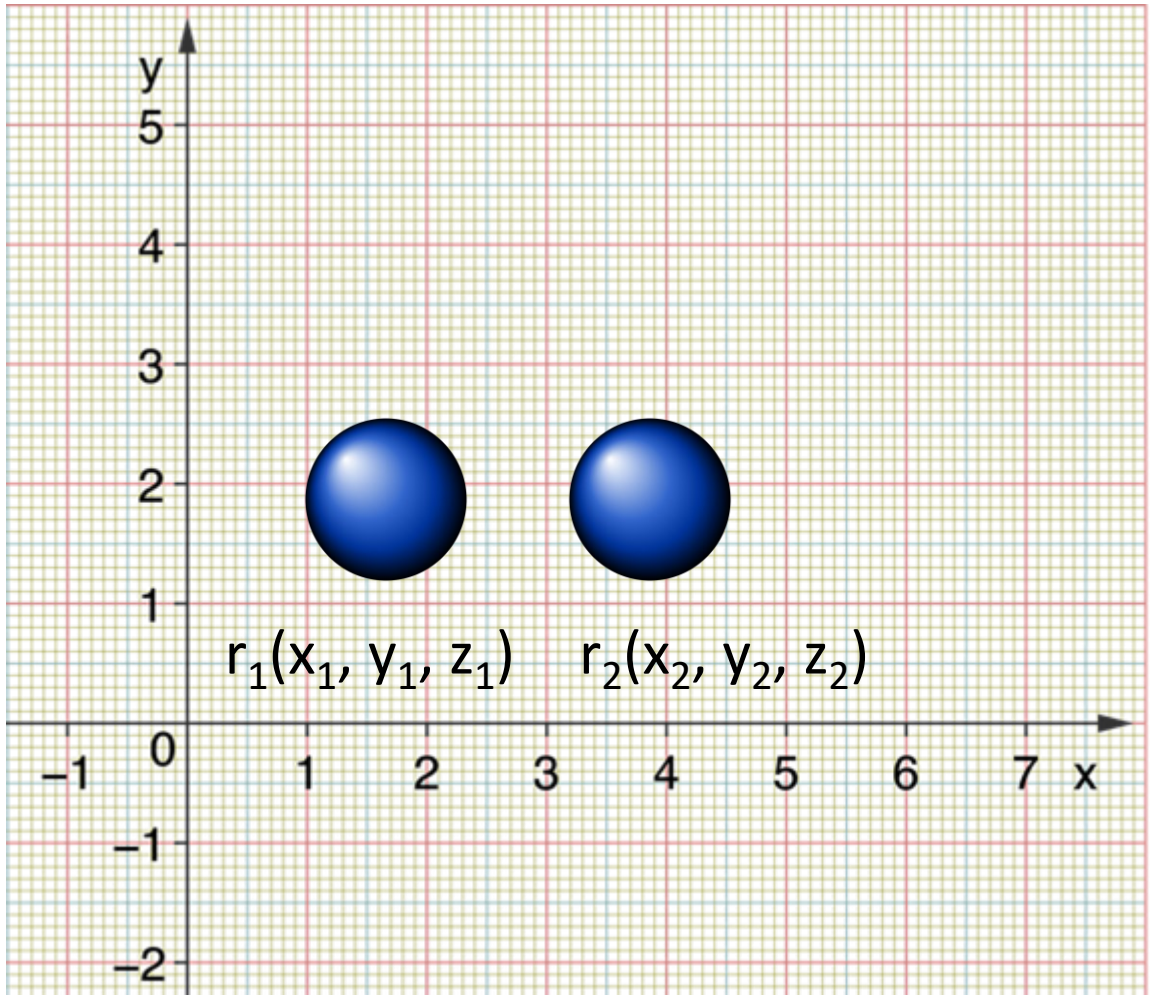
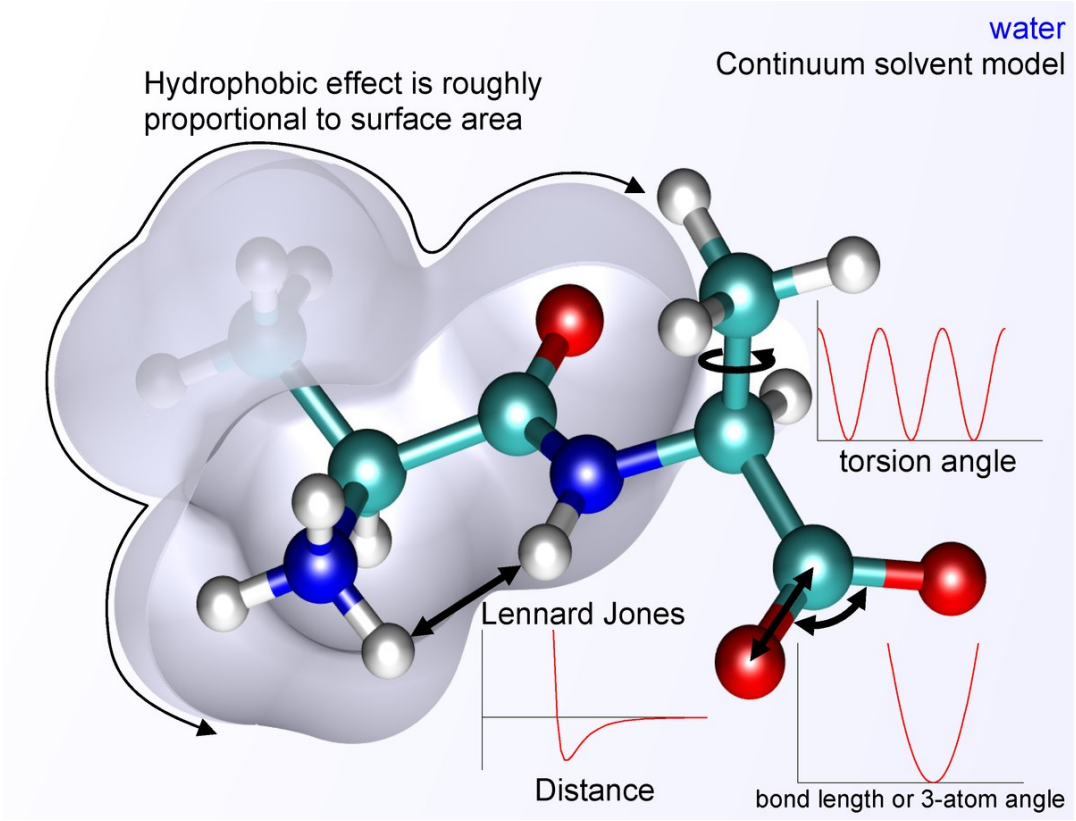


- Input initial conditions:
  - Position: r

Theories Molecular dynamics (MD) simulation

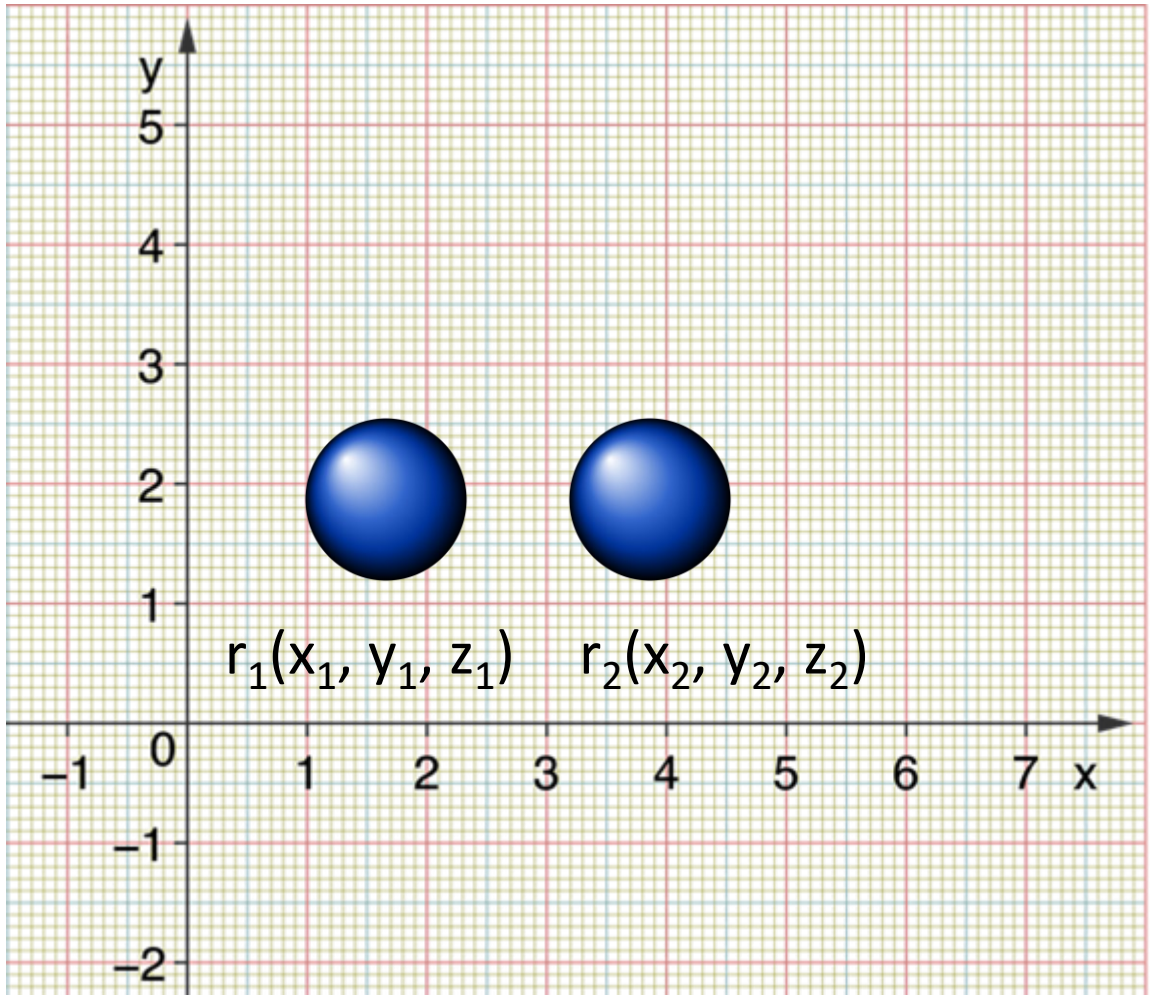


- Input initial conditions:
  - Position: r
  - Potential interaction: V (Force field)



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Theories Molecular dynamics (MD) simulation



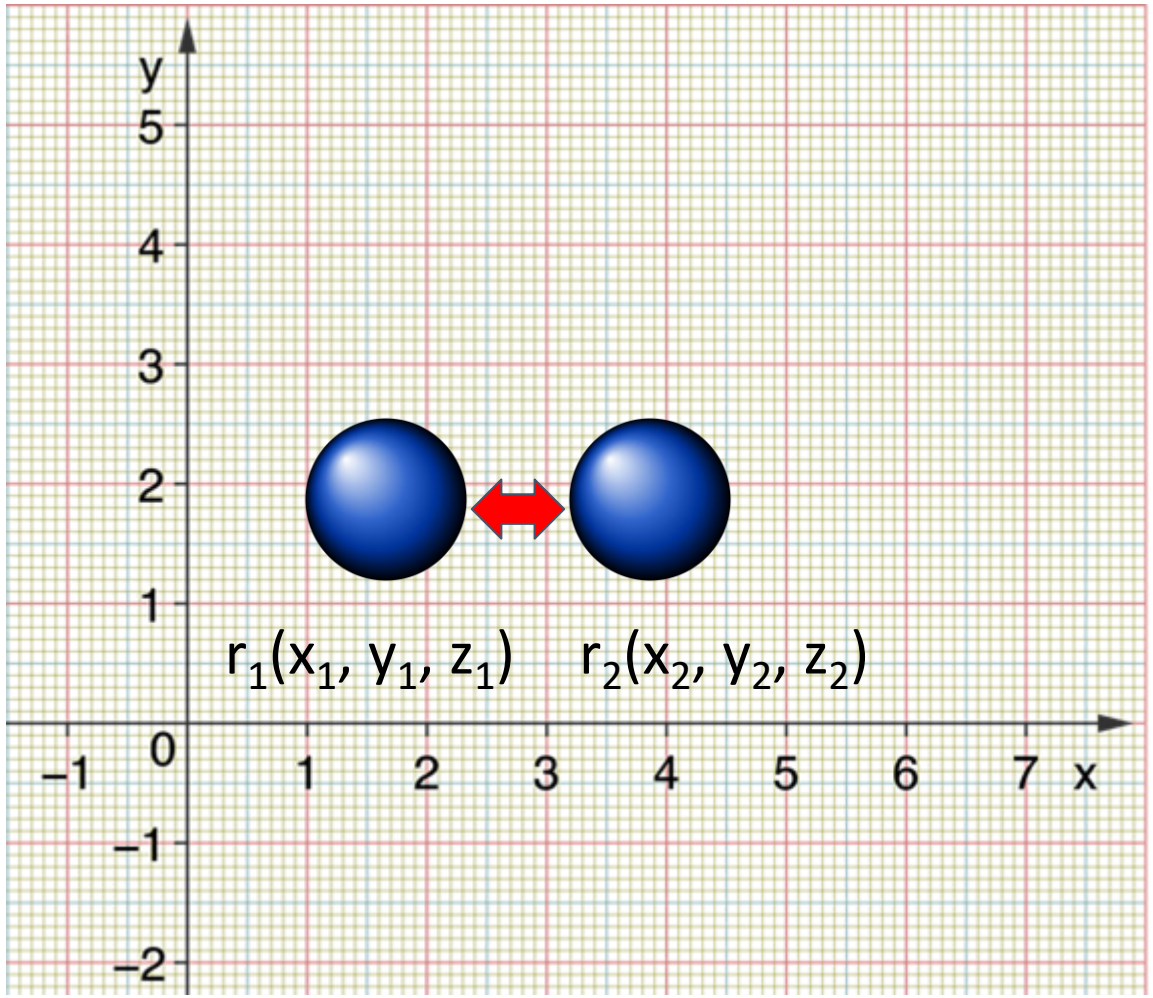
- Input initial conditions:
  - Position: r
  - Potential interaction: V (Force field)

$$E_{total} = E_{bonds} + E_{angles} + E_{dihedrals} + E_{vdw+electrostatic}$$

$$= \sum_{bonds} K_r(r - r_{eq})^2 + \sum_{angles} K_\theta(\theta - \theta_{eq})^2 +$$

$$\sum_{dihedrals} \frac{V_n}{2}[1 + \cos(n\phi - \gamma)] + \sum_{i<j} [\frac{A_{ij}}{R_{ij}^{12}} + \frac{B_{ij}}{R_{ij}^{6}} + \frac{q_i q_j}{\epsilon R_{ij}}]$$

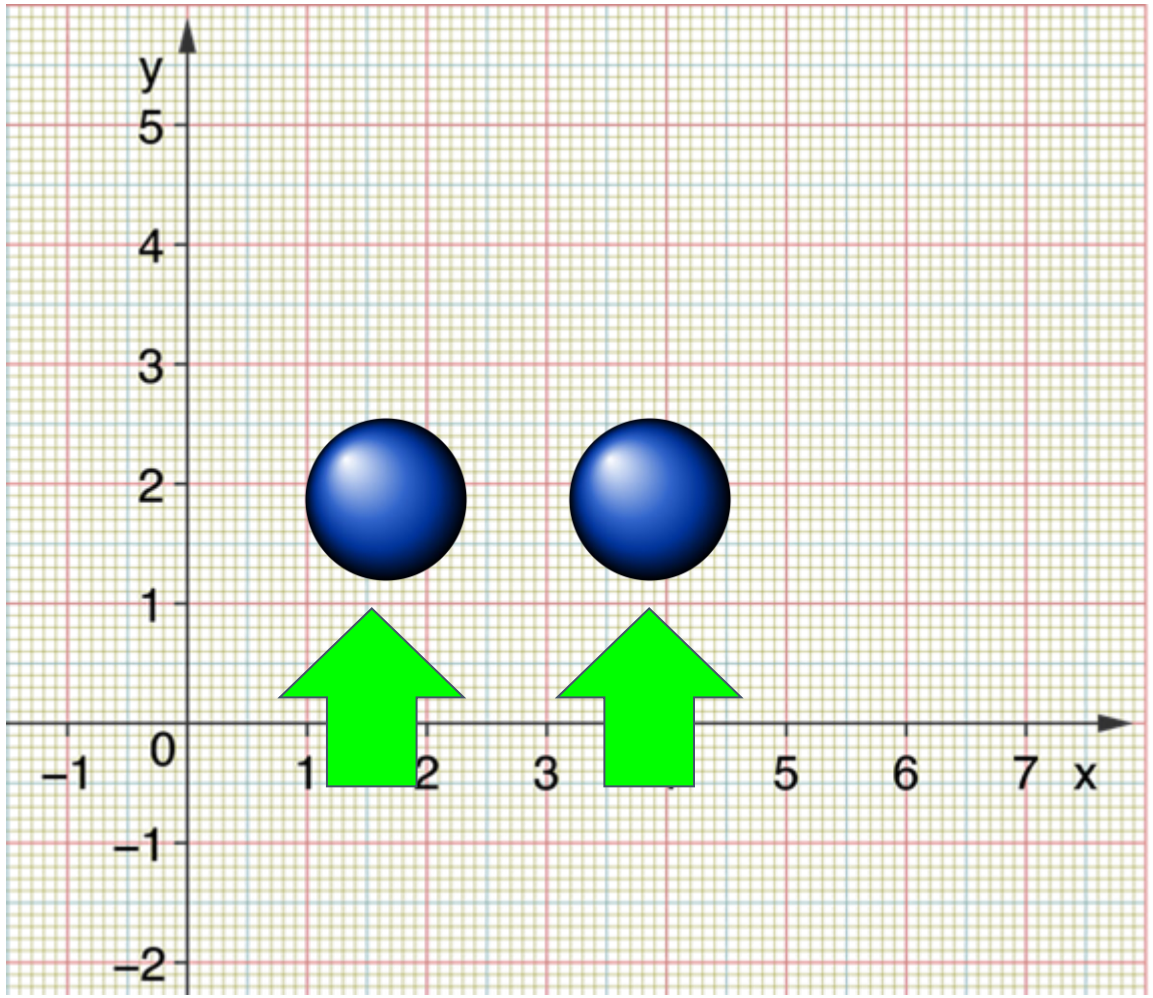
Theories Molecular dynamics (MD) simulation



- Input initial conditions:
  - Position: r
  - Potential interaction: V (Force field)
- Calculate forces: Newton's 2nd law

$$F = ma = m\frac{\partial^2 r}{\partial t^2} = -\frac{\partial V}{\partial r}$$

Theories Molecular dynamics (MD) simulation

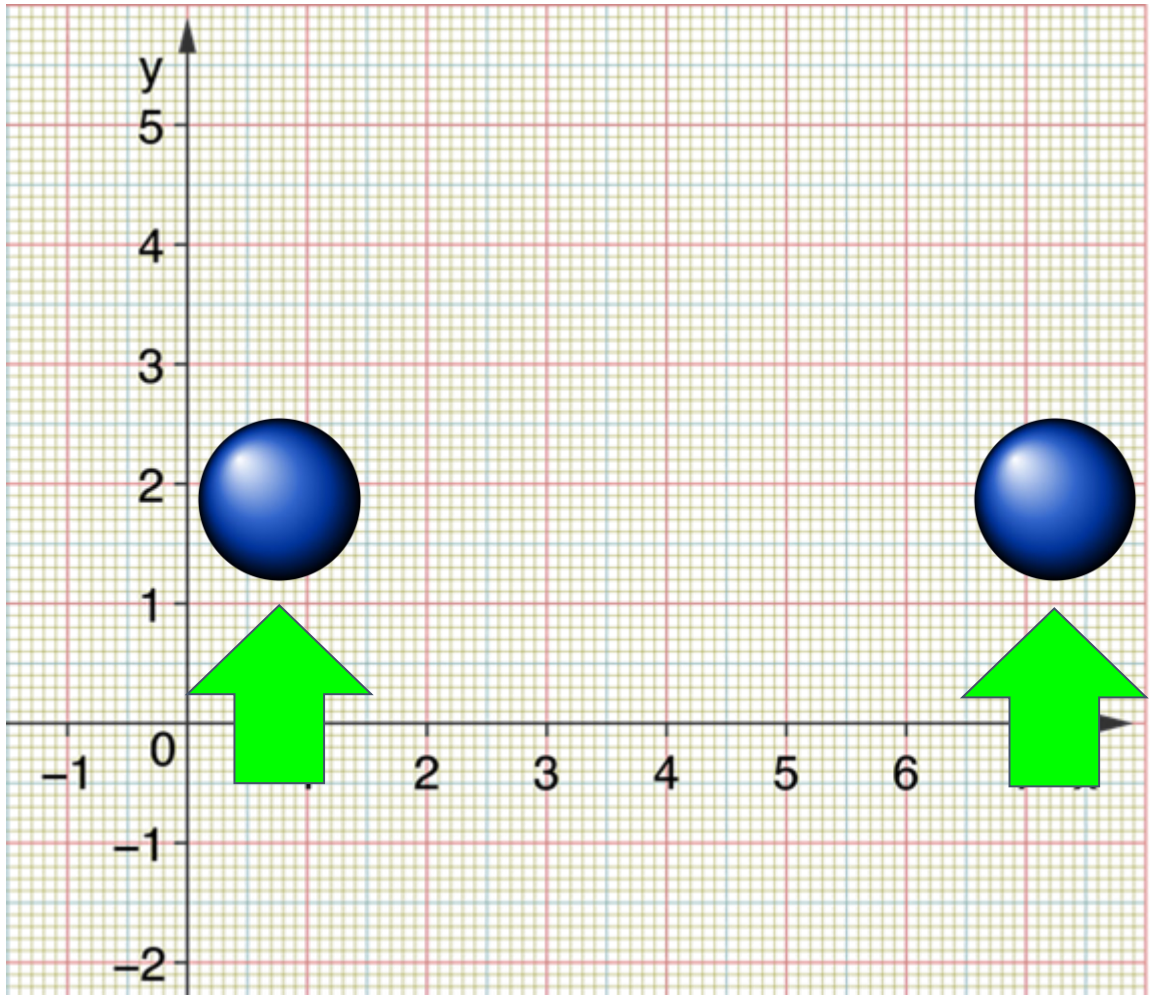


- Input initial conditions:
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  - Potential interaction: V (Force field)
- Calculate forces: Newton's 2nd law

$$F = ma = m\frac{\partial^2 r}{\partial t^2} = -\frac{\partial V}{\partial r}$$

- Update configurations

Theories Molecular dynamics (MD) simulation

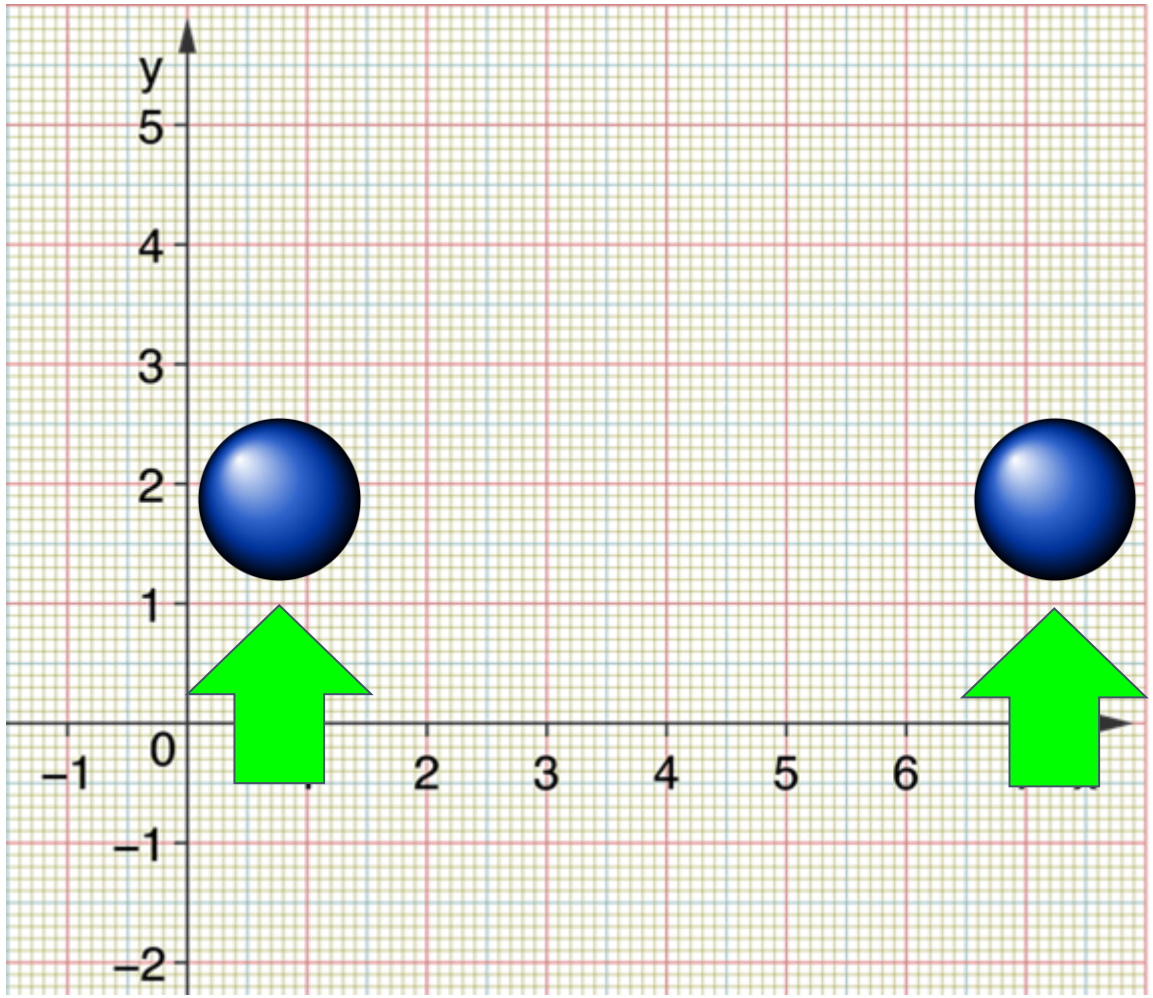


- Input initial conditions:
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$$F = ma = m\frac{\partial^2 r}{\partial t^2} = -\frac{\partial V}{\partial r}$$

- Update configurations

$$\frac{\mathrm{d}^2 r}{\mathrm{d}t^2} = \frac{F}{m}$$

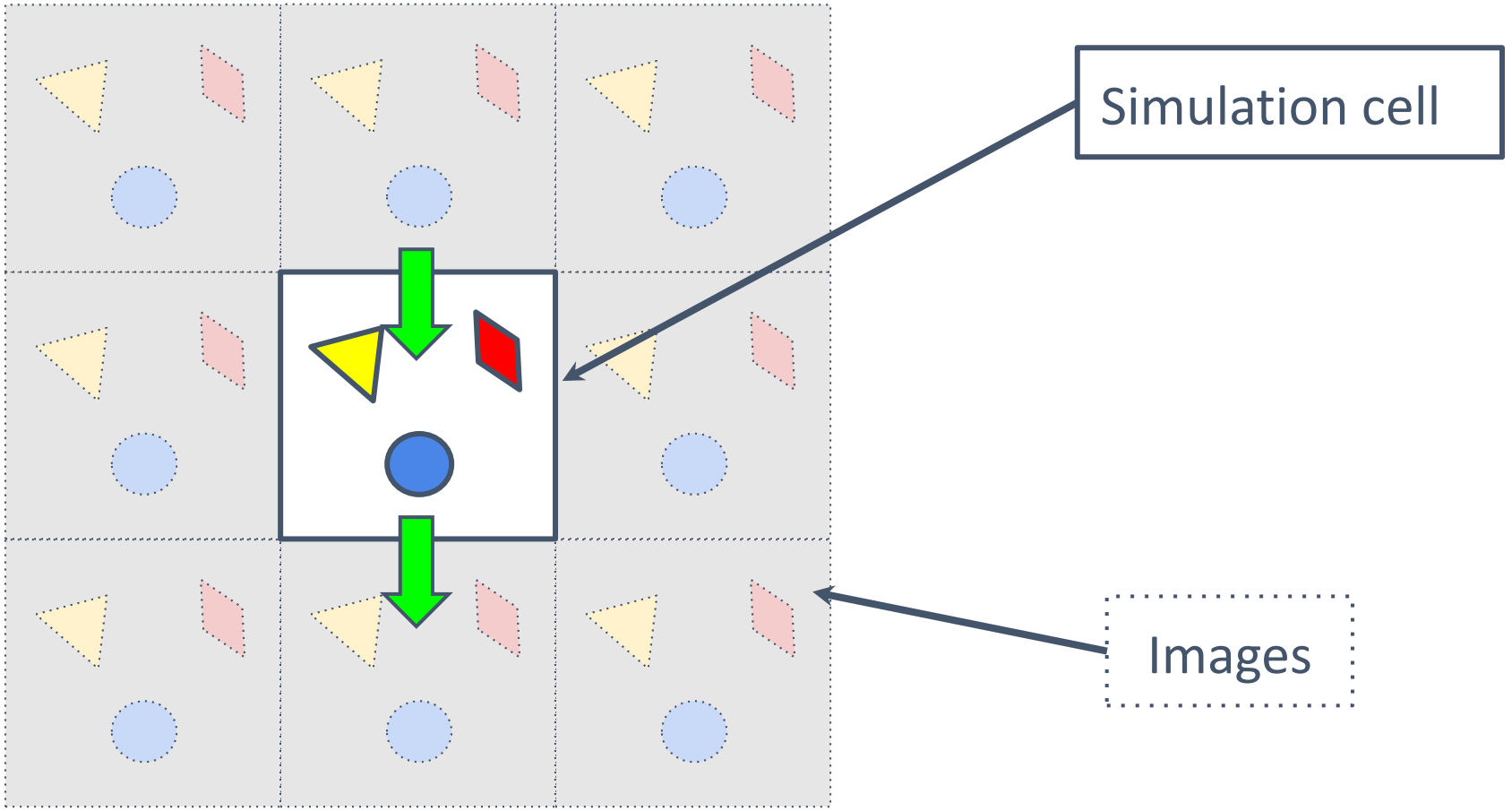


- Input initial conditions:
  - Position: r
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- Calculate forces: Newton's 2nd law

$$F = ma = m\frac{\partial^2 r}{\partial t^2} = -\frac{\partial V}{\partial r}$$

- Update configurations

$$\frac{\mathrm{d}^2 r}{\mathrm{d}t^2} = \frac{F}{m}$$


Simulation cell
Images